

Web-Based Attendance Monitoring System with Facial Recognition and Irregularity Detection Using FaceNet, Isolation Forest, and Decision Tree

Dino F. Agito

CS3A

CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

Background of the Problem

In recent years, technological progress has had a major impact across multiple sectors, particularly in education and organizational management. A growing number of schools and workplaces now use digital attendance systems to streamline the process of recording, storing, and accessing attendance information. Such systems help minimize manual tasks, make records easier to retrieve, and enable quicker administrative reporting. However, despite these improvements, most current attendance solutions remain focused on collecting and storing data, rather than offering advanced, intelligent monitoring capabilities.

As a result, administrators are often required to manually review attendance logs to identify irregularities such as repeated lateness, duplicate rapid check-ins, unusual check-in times, missing time-out records, unrealistic attendance durations, and sudden changes in attendance patterns. This approach is labor-intensive, susceptible to mistakes, and can slow down necessary corrective actions, potentially undermining discipline, responsibility, and overall productivity. Research indicates that machine learning methods outperform manual review in uncovering subtle patterns and anomalies within complex data (Sarker, 2021; Pang et al., 2021).

Although web-based attendance systems have improved data management, they remain limited in automatically analyzing attendance behavior. In many institutions, including educational and workplace settings, attendance monitoring still relies on



manual verification to determine whether recorded patterns are normal or suspicious. This indicates that while digital systems improve record-keeping, they do not fully address the need for intelligent analysis and real-time decision support.

Recent advancements in machine learning and deep learning provide opportunities to enhance attendance monitoring systems. In particular, facial recognition has emerged as a reliable method for identity verification, as deep learning models can accurately extract and compare facial features for automated identification (Wang & Deng, 2021). Furthermore, convolutional neural networks (CNNs) have proven especially adept at capturing relevant visual features, making them well-suited for recognition tasks (Alzubaidi et al., 2021). Moreover, anomaly detection methods like Isolation Forest can spot irregular patterns without relying on extensive labeled datasets, increasing their practicality in real-world applications (Pang et al., 2021).

With the integration of FaceNet, a deep learning facial recognition system based on convolutional neural networks, identity verification during attendance tracking can be greatly enhanced. Concurrently, the Isolation Forest algorithm helps identify unusual attendance patterns, while a Decision Tree classifier organizes attendance data into clear categories like regular, late, or at-risk. Together, these methods shift the focus from basic attendance logging to more advanced, intelligent monitoring and analysis. The practical effectiveness of face recognition technologies is further supported by the National Institute of Standards and Technology (NIST, 2024), which highlights their continued relevance in identity verification applications.

However, despite these advances, current research and systems typically concentrate on isolated methods, such as facial recognition, anomaly detection, or classification—without effectively combining them into a cohesive, web-based



attendance solution. Many existing implementations prioritize data logging and storage, offering little support for real-time validation or the automatic identification of unusual attendance patterns in real-world settings. This gap highlights the need for a system that integrates multiple machine learning techniques to bridge web-based attendance management and intelligent data analysis.

Therefore, this study proposes the development of a web-based attendance monitoring system integrated with facial recognition and irregularity detection using FaceNet, Isolation Forest, and Decision Tree. The system aims to improve identity verification, detect irregular attendance patterns at an early stage, reduce manual monitoring, and support faster and more reliable administrative decision-making.

Statement of the Problem

Web-based attendance monitoring systems are commonly used in schools and workplaces to streamline the recording, storing, and accessing of attendance records. While these systems simplify log management and report generation, many still focus primarily on data storage and lack advanced capabilities for analyzing attendance behavior. As a result, administrators often still need to manually inspect raw logs to identify irregularities such as repeated lateness, duplicate check-ins, unusual check-in times, unrealistic attendance durations, and sudden changes in attendance patterns. This process is time-consuming, may delay intervention, and can still lead to inconsistencies or human error in decision-making. In addition, identity verification remains a challenge in some attendance systems, especially when manual checking or basic authentication methods are used. With the use of FaceNet for facial recognition,



Isolation Forest for anomaly detection, and Decision Tree for behavior classification, it becomes possible to develop a more intelligent web-based attendance monitoring system that not only records attendance but also detects irregularities and classifies behavior more effectively. This study therefore assesses how effectively a web-based attendance system enhanced with FaceNet, Isolation Forest, and Decision Tree can improve both monitoring and analysis in educational or organizational environments.

Specifically, this will seek to answer the following questions:

- How effective are Isolation Forest and Decision Tree, integrated with FaceNet in a web-based attendance system, at detecting irregularities and classifying behaviors compared to manual methods?
- What accuracy, precision, and recall does the Isolation Forest model achieve in detecting anomalies like duplicate check-ins, unusual times, and unrealistic durations using train-test split and confusion matrix evaluation?
- How well does the Decision Tree classify attendance into regular, late, and at-risk categories, measured by confusion matrix metrics on institutional attendance data?
- Does FaceNet provide reliable employee identification for attendance logging, evaluated through recognition accuracy on collected face image samples?
- To what extent does the integrated system improve monitoring efficiency, usability (SUS), and quality (ISO 25010) over traditional approaches in a school/workplace setting?



Objectives of the Study

The general objective of this study is to develop and evaluate a web-based attendance monitoring system integrated with FaceNet, Isolation Forest, and Decision Tree to improve attendance verification, irregularity detection, and behavior classification in school and workplace settings.

Specifically, this study aims to:

- To implement an integrated web-based attendance monitoring system using FaceNet, Isolation Forest, and Decision Tree.
- To design, train, and evaluate Isolation Forest model performance using accuracy, precision, and recall metrics through train-test split and confusion matrix for detecting anomalies such as duplicate check-ins, unusual times, and unrealistic durations.
- To design, train, and evaluate Decision Tree model performance using confusion matrix metrics to classify attendance behaviors into regular, late, and at-risk categories on institutional attendance data.
- To implement and evaluate FaceNet facial recognition reliability for employee identification during attendance logging using recognition accuracy on collected face image samples.
- To assess improvements in monitoring efficiency, usability (SUS), and system quality (ISO 25010) achieved by the integrated system compared to traditional approaches in school and workplace settings.



Significance of the Study

This study holds substantial practical and academic importance by addressing critical gaps in attendance monitoring through machine learning integration.

- **Administrators:** benefit from automated detection of attendance irregularities such as duplicate check-ins, unusual times, and unrealistic durations, enabling faster and more objective interventions compared to manual log reviews.
- **Instructors and Supervisors:** save time by eliminating manual analysis of large attendance datasets, receiving real-time alerts and interpretable classifications (regular, late, at-risk) for timely disciplinary actions.
- **Students and Employees:** experience fairer monitoring through reliable FaceNet facial recognition that prevents proxy logging while maintaining privacy via biometric backup verification.
- **Institutions:** strengthen attendance policy enforcement with data-driven insights, reducing administrative workload and supporting resource allocation decisions based on accurate behavioral patterns.
- **Future Researchers:** gain a validated framework combining Isolation Forest anomaly detection, Decision Tree classification, and FaceNet integration, serving as a scalable model for ML-enhanced institutional monitoring systems.

Scope and Limitations

This study includes the design, development, and evaluation of a web-based attendance monitoring system integrated with a facial recognition and irregularity recognition module incorporating FaceNet, Isolation Forest, and Decision Tree



algorithms. Student attendance data is captured by RFID technology, and attendance for employees is done through facial recognition based on FaceNet with biometrics as a secondary authentication approach. The Isolation Forest algorithm is used to identify anomalies, and decision trees are used to classify user behavior as normal, late, and at risk. RFID is implemented for students to support fast processing and scalability in large populations, while facial recognition is applied to employees to ensure higher identity verification accuracy.

This study is limited to the development and evaluation of web-based attendance monitoring systems based on the institutional data gathered during the study period . The model's performance relies on the attendance logs and employee facial image samples being of high quality, complete and representative. Irregularity detection is confined to pre-defined patterns and engineered features such as duplicate rapid check-ins, unusual check-in times, missing time-outs, unrealistic durations and repeated lateness. Facial recognition is only for employees and student attendance is logged only with RFID. The system provides alerts and behavior classifications as a decision support tool, but the final validation and disciplinary actions are left to the discretion of the administrators. The evaluation results are specific to the context of the participating institution and may need to be recalibrated when applied to different schedules, policies, hardware configurations or operating environments. Moreover, operational issues such as camera quality, lighting conditions, network interruptions, and device malfunctions can impact data capture and model reliability. The study does not include deployment across institutions over the long term, development of mobile-native applications, or a full assessment of legal compliance aside from the consideration of institutional policies.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter examines related literature and studies essential to the development of a web-based attendance monitoring system integrating facial recognition and machine learning-based irregularity detection. It explores the role of attendance systems with facial recognition technologies, limitations of Isolation Forest for anomaly detection in attendance records, challenges in behavioral pattern analysis for irregularities like duplicate check-ins and unrealistic durations, and classification approaches using Decision Tree algorithms. The reviewed sources identify critical technical gaps and practical limitations that establish the conceptual framework and justify the proposed system's design, algorithms, and evaluation methodology.

General Role of Attendance Systems with Facial Recognition and Anomaly Detection

Facial recognition attendance systems have been increasingly adopted in educational and organizational settings to automate attendance verification and reduce proxy attendance. These systems commonly utilize algorithms such as FaceNet embeddings, which achieve high accuracy under controlled conditions but exhibit limitations in real-world deployment. FaceNet-based systems are sensitive to environmental factors including poor lighting, variations in facial angles, occlusions, and changes in appearance over time, which can significantly affect recognition performance (Rajkumar, 2024; Wang & Deng, 2021). In addition, existing attendance tracking systems often lack real-time anomaly detection and context awareness, limiting their ability to



promptly identify irregularities such as duplicate check-ins or unrealistic attendance durations. Prior studies emphasize that while machine learning techniques are effective in pattern recognition, many practical systems still lack integrated anomaly detection capabilities (Pang et al., 2021). This gap highlights the need for integrated systems that combine robust facial recognition with unsupervised anomaly detection techniques such as Isolation Forest, which can identify unusual patterns without requiring large labeled datasets. Such integration addresses both the environmental limitations of facial recognition and the need for scalable, real-time detection of attendance irregularities in practical deployments. This indicates that a more effective attendance system should integrate facial recognition with anomaly detection techniques to enable both accurate identification and real-time behavioral analysis in a unified platform.

Challenges in Anomaly Detection for Attendance Monitoring

Isolation Forest has shown its effectiveness in unsupervised anomaly detection. However, there are significant behavioral challenges for existing web-based attendance systems in detecting proxy attendance patterns. Singh et al. (2026) developed a behavioral anomaly detection framework specifically targeting proxy attendance in web-based systems, revealing that current approaches struggle with distinguishing legitimate rapid check-ins from fraudulent multiple-user access within short time windows. The analysis shows the false positive rate goes over 15% when trying to determine the presence of duplicate tapping behaviors, caused by legitimate group arrivals, and that feature engineering for temporal patterns is often insufficient, resulting in unreliable session durations or sudden changes in attendance behavior. These findings are consistent with broader anomaly detection studies, which highlight that



unsupervised models may struggle to differentiate between normal clustered behavior and true anomalies in real-world datasets (Pang et al., 2021). These limitations underscore the need for hybrid approaches combining Isolation Forest's anomaly scoring with Decision Tree's interpretable classification rules, particularly for institutional datasets with limited labeled examples. This limitation highlights the need to combine anomaly detection with classification techniques to improve accuracy and reduce false positives in real-world attendance monitoring systems.

Isolation Forest Limitations in Attendance Anomaly Detection

While Isolation Forest demonstrates strong potential for detecting anomalies in RFID-based attendance systems, recent implementations highlight several practical limitations in real-world deployment. Nalupa et al. (2024) reported that their RFID attendance monitoring system achieved high performance in detecting suspicious attendance patterns; however, the system incorporated fingerprint verification for cases flagged as anomalies, indicating the presence of false positives arising from legitimate group arrivals and rapid check-in behaviors. This suggests that Isolation Forest may be sensitive to time-based clustering in institutional datasets, where normal synchronized entries can be misclassified as fraudulent activity. Similar observations have been noted in anomaly detection research, where unsupervised models tend to misinterpret dense or patterned data as anomalies due to lack of contextual understanding (Pang et al., 2021). Such limitations highlight challenges in distinguishing genuine attendance behavior from anomalies in real-world environments. These challenges suggest that anomaly detection alone is insufficient, and should be complemented with additional



methods such as classification models to improve reliability and interpretation of detected irregularities.

Decision Tree Classification Limitations in Attendance Systems

Decision Tree models have been widely used in attendance systems due to their simplicity and interpretability, but recent studies have shown that their performance has limitations when dealing with real-world attendance data. One key challenge is their sensitivity to imbalanced datasets where the number of normal attendance records is much larger than the minority classes, which in attendance context may be irregular behaviors such as absenteeism or tardiness. This imbalance causes Decision Trees to favor majority patterns, which limits their ability to accurately detect less frequent yet significant attendance anomalies (Faiz et al., 2026). Furthermore, Decision Trees tend to be unstable and overfit, particularly when dealing with complex or overlapping behavioral characteristics, where small changes in the training data can lead to different tree structures and inconsistent classification rules (García-Gil et al., 2024). This makes it challenging to accurately simulate real-world attendance behaviors in attendance monitoring systems, where there is often an overlap in patterns between normal and irregular users. However, the limitations of standalone Decision Tree in attendance classification tasks reduce its reliability, thereby supporting the need for hybrid approaches that include anomaly detection techniques such as Isolation Forest to improve detection accuracy and reliability. These limitations indicate that Decision Trees alone may not be sufficient for accurate attendance behavior classification, and should be integrated with anomaly detection techniques such as Isolation Forest to improve overall system performance.

Local Facial Recognition Limitations in Philippine Institutions

Facial recognition attendance systems have been deployed in several educational institutions in the Philippines to enhance efficiency and minimize manual attendance procedures. However, local studies show several practical constraints in their deployment. Facial recognition systems are extremely sensitive to environmental conditions like lighting variations, camera positioning, and hardware limitations, which dramatically impact recognition accuracy in real-world settings, according to Domingo & Ladia (2024). In addition, Makigod et al. (2024) found that system performance is influenced by infrastructure constraints, including limited processing capabilities and inconsistent network connectivity, which are common challenges in local implementations. Furthermore, local studies indicate concerns about user privacy, data security, and the significance of adequate user training for the reliable use of the system. Facial recognition also improves the automation of attendance monitoring but does not completely eliminate errors such as misidentification or failure to recognize faces under different environmental conditions. There is a need for constant improvement and validation of the system. These constraints highlight the importance of robust, context-sensitive attendance systems that combine facial recognition with additional methods to improve reliability and flexibility in Philippine institutional environments. These findings emphasize the need for an integrated attendance system that combines facial recognition with complementary techniques to address environmental and infrastructure-related limitations in real-world institutional settings.

Synthesis

The reviewed literature highlights the growing use of attendance monitoring systems enhanced by facial recognition and machine learning techniques. Studies



indicates that facial recognition tools, especially FaceNet, enhance identity confirmation and help minimize instances of proxy attendance. Nevertheless, the effectiveness of these systems can be notably influenced by environmental factors like lighting conditions, camera resolution, and variations in facial appearance, potentially compromising accuracy in practical applications.

In terms of anomaly detection, Isolation Forest is widely recognized for its ability to detect unusual patterns without requiring labeled data. Despite this advantage, it struggles with distinguishing legitimate behaviors—such as group arrivals—from actual anomalies, often resulting in false positives. Similarly, Decision Tree algorithms are valued for their interpretability and simplicity in classifying behaviors, but their performance is limited when handling imbalanced datasets and complex behavioral overlaps, which may lead to inaccurate classifications.

Local studies in the Philippine context further emphasize challenges such as infrastructure limitations, inconsistent network connectivity, and privacy concerns, all of which affect the reliability of attendance systems. These findings indicate that no single approach is sufficient to fully address attendance monitoring issues.

Therefore, integrating FaceNet for identification, Isolation Forest for anomaly detection, and Decision Tree for behavior classification provides a more comprehensive solution. This hybrid approach addresses the limitations of individual methods by combining accuracy, scalability, and interpretability, ultimately improving the efficiency, reliability, and practicality of attendance monitoring systems in institutional settings.

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